TÉCNICO LISBOA

AVTM 26/27

Assignment 1

The evaluation of this assignment's tasks as well as the tasks of Exercise 2 and Exercise 3 will be held in the first lab class of week 4, **September 30th,** and corresponds to **60%** of the total Laboratory grade. The required tasks are listed below; however, it is expected that each group enriches the game with other elements and features to make it more attractive.

Tasks

The tasks of Exercises 2 and 3 will be evaluated with **17 points**: (see below the grading)

The tasks of this assignment are:

- to implement the planar reflections and planar shadows **(2 points)** by using the blending and stencil mechanisms as explained in the theoretical class.
- Extra work (**1 point**). For instance:
  - Bump-mapping
  - Skybox and Environment cube mapping

Regarding the above tasks, the students can base their development on attached ReflectionsAndPlanarShadows.zip and AdvancedTexturing.zip demo files.

Grading (Exercises 2 and 3)

1 – Graphic modelling (include OBJ support) **[2.5 points]**

2 – Cameras **[1.5 points]**

3 - Game elements' movement **[2 points]**

4 - Lighting of the scene **[3 points]**

a. Directional light **[0.5 point]**
b. Six point lights **[1 point]**
c. Spot lights **[1.5 points]**

5 – Collision detection **[2 points]**

6 – Texture mapping **[1 point]**

7 – Transparency **[1 point]**

8 – Fog effect **[1 point]**

9 – HUD and Pause **[1 point]**

10 – Transparencies **[1 point]**

11 – Rearview **[1 point]**

Notes

1. The Groups should upload their source files in the Fénix System at the end of their evaluation.

2. The Groups should deliver by email (jap@inesc-id.pt) , until **October 7th**, a **technical report up to 6 pages**. The report does not have to fully explain the techniques. It should rather inform, for each feature of the game, which techniques were used and how, illustrating with code excerpts, drawings and/or images. Disclose the IA tools used in the Assignment.

3. The **total grade** of this assignment will be calculated by coding quality (95%) and technical report (5%).